

Day 15 — Sat 19 Sep

Countdown: 7 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🎯 GK — Economy

Source: General/03_GK_CAPSULE.md

5.4 The Union Budget

- Presented by the **Finance Minister**, usually on **1 February**.
- **Article 112** — the "Annual Financial Statement."
- **Revenue Budget** (recurring income/expenditure) vs **Capital Budget** (assets & liabilities). **Fiscal deficit** = total expenditure – total receipts (excluding borrowings).

5.7 Major government schemes (name → purpose)

Scheme	Purpose
MGNREGA	Guarantees 100 days of wage employment (rural)
PM Jan Dhan Yojana	Financial inclusion — bank accounts for all
Ayushman Bharat (PM-JAY)	Health insurance cover for poor families
Digital India	Digital infrastructure & e-governance
Swachh Bharat Mission	Sanitation, toilets, cleanliness
PM Awas Yojana	Housing for all
Jal Jeevan Mission	Tap water to every rural household
Skill India / Startup India	Skilling & entrepreneurship
Act East Policy	Engagement with Southeast Asia — directly relevant to the NE & Nagaland's Myanmar border

⚠ verify latest outlays/coverage numbers before quoting any figure.

Economy — likely exam questions

- [5] Distinguish between fiscal policy and monetary policy.
- [5] What is GST? Explain CGST, SGST and IGST.
- [5] Differentiate between GDP and GNP.
- [10] Explain the functions of the Reserve Bank of India.

Economy — MCQ traps (with answers)

Answers

1. GDP + net income from abroad = ? → ****GNP**** 2. GST came into force on? → ****1 July 2017**** 3. Repo rate

is set by? → **RBI (Monetary Policy Committee)** 4. The Union Budget is presented under which article? → **Article 112** 5. NITI Aayog replaced? → **the Planning Commission (2015)** 6. Monetary policy is controlled by? → **RBI**, fiscal policy by the **Government**

SECTION 6 — GEOGRAPHY

● Core — OSI & TCP/IP

Source: `_Shared/CORE_05_ComputerNetworks.md`

6. Connectivity devices — mapped to OSI layers

Concept

Every internetworking device does exactly one thing: it **reads a header and makes a decision**. The *layer* a device operates at is simply **the highest header it reads**. A repeater reads nothing (it just amplifies) → Layer 1. A switch reads the MAC header → Layer 2. A router reads the IP header → Layer 3. Once you internalise that rule, the whole table becomes derivable instead of memorised.

The master device table

Device	OSI layer	Reads / forwards on	What it does	Collision domains	Broadcast domains
Repeater	1 — Physical	Nothing (bit stream)	Regenerates and retimes a weakened digital signal to extend distance. Not merely an amplifier — an amplifier would boost the noise too; a repeater reconstructs the bits	1 (one shared domain)	1
Hub (multiport repeater)	1 — Physical	Nothing	Receives a frame on one port and floods it out of every other port . No filtering, no addresses. Half-duplex, CSMA/CD . Obsolete	1 — all ports share one collision domain	1
Bridge	2 — Data Link	MAC address	Connects two LAN segments and filters : learns which MACs are on which side and forwards a frame only if the destination is on the other side. Software-based, few ports	One per port	1
Switch (multiport bridge)	2 — Data Link (L3 switches also do 3)	MAC address	Builds a MAC address table (CAM table) by learning source addresses; forwards each frame only to the destination port (unicast), floods only unknown/broadcast/multicast. Hardware (ASIC) based, full duplex	One per port — this is the key advantage	1 (one per VLAN)

Router	3 — Network	IP address	Connects different networks ; makes forwarding decisions from a routing table ; determines the best path ; performs fragmentation , TTL decrement; blocks broadcasts	One per port	One per port — routers break up broadcast domains
Gateway	All 7 (up to Application)	Everything — translates	A protocol converter : joins two networks using completely different protocol architectures (e.g. TCP/IP ↔ SNA, or an email gateway between SMTP and X.400). Slowest device; usually software on a server	—	—
Modem	1 — Physical	—	Modulator–demodulator : converts digital → analogue for transmission over an analogue line and back. Enables digital data over the telephone network	—	—
NIC (network adapter)	1 and 2	—	Physical interface; holds the burned-in MAC address ; frames/deframes	—	—
Access Point (AP)	2	MAC	Bridges a wireless (802.11) segment to a wired LAN; uses CSMA/CA	1 (shared radio)	1
Brouter	2 and 3	Both	Routes routable protocols, bridges non-routable ones	—	—
Proxy server	7 — Application	Application data	Sits between clients and the Internet: caches, filters content, hides internal addresses	—	—
Firewall	3–7 (depends on type)	Headers and/or payload	Filters traffic against a rule set	—	—

The two sentences that earn marks

- **"A switch breaks up collision domains; a router breaks up both collision domains and broadcast domains."** This one sentence answers a whole family of questions. Corollary: a hub breaks up neither.
- **"A hub floods, a switch forwards, a router routes."**

Worked example: A 24-port switch has **24 collision domains** and **1 broadcast domain**. A 24-port hub has **1** collision domain and **1** broadcast domain. A router with 4 interfaces gives **4** broadcast domains. If three 8-port hubs are each plugged into one port of a switch, you have 3 collision domains (one per hub) plus the remaining switch ports, and still one broadcast domain.

Hub vs switch vs router — the standard 5-mark table:

	Hub	Switch	Router
OSI layer	1	2	3
Address used	None	MAC	IP

Forwarding	Broadcast to all ports	Unicast to the correct port	Best path from routing table
Table maintained	None	MAC/CAM table	Routing table
Duplex	Half	Full	Full
Collision domains	1	Per port	Per port
Broadcast domains	1	1 (per VLAN)	Per port
Connects	Devices in one LAN	Devices in one LAN	Different networks
Speed/cost	Cheapest, slowest	Fast, moderate	Slower per packet, costliest

Switching methods inside a switch (a nice extra for 10 marks):

Method	Behaviour	Latency	Error checking
Store-and-forward	Receives the whole frame, verifies the CRC , then forwards	Highest	Full — drops bad frames
Cut-through	Reads only the first 6 bytes (destination MAC) and starts forwarding	Lowest	None — forwards corrupt frames too
Fragment-free	Reads the first 64 bytes (where most collisions show) then forwards	Medium	Partial

Likely exam questions

- **[5]** Differentiate between a hub and a switch.
- **[5]** Explain the function of a router. At which OSI layer does it operate?
- **[5]** What is a gateway? How does it differ from a router?
- **[5]** What is a modem? Explain modulation and demodulation.
- **[5]** Define collision domain and broadcast domain. How many of each does a 16-port switch create?
- **[10]** Explain the various network connectivity devices — repeater, hub, bridge, switch, router, gateway and modem — mapping each to its OSI layer and stating its function.
- **[15]** Discuss internetworking devices in detail. Explain how each operates, map them to the OSI model, explain collision and broadcast domains with examples, and compare hub, switch and router in a table.

MCQ traps

Trap	Correct answer
"A switch operates at layer?"	2 (Data Link). Router = 3 . Hub/repeater = 1 .
"Which device breaks broadcast domains?"	Router (not switch).
"Which device works at all seven layers?"	Gateway .
"A bridge uses which address?"	MAC .
"Multiport repeater = ?"	Hub . Multiport bridge = switch .
"Which device does fragmentation?"	Router .
"A hub is full-duplex?"	No — half-duplex , with collisions.
"Modem converts?"	Digital ↔ analogue.

7. The OSI reference model — in full

Concept

The problem OSI solves is this: building a network in one piece is impossible, because a change to the cabling would force a change to your email program. So the ISO (International Organization for Standardization) published, in **1984**, the **Open Systems Interconnection** model: a **seven-layer** architecture in which each layer performs a well-defined function, uses only the services of the layer immediately below, and provides services only to the layer immediately above. Replace the physical medium and only Layer 1 changes.

Two vocabulary items to state early:

- **Peer-to-peer communication**: layer *n* on the sender logically talks to layer *n* on the receiver, using a **protocol**. Physically, of course, the data goes all the way down, across, and back up.
- **Encapsulation**: each layer adds its own **header** (and Layer 2 adds a **trailer**) to the unit received from above. That growing unit is the layer's **PDU** (Protocol Data Unit). The receiver strips the headers in reverse — **de-encapsulation**.

Mnemonics — learn one in each direction:

- Layer 7 → 1: **All People Seem To Need Data Processing**
- Layer 1 → 7: **Please Do Not Throw Sausage Pizza Away**

Diagram to hand-draw — the OSI stack with encapsulation

Draw **two vertical stacks of seven boxes** (sender on the left, receiver on the right), numbered 7 at the top down to 1 at the bottom. Join the two Layer-1 boxes with a horizontal line labelled "**physical medium**". Draw **dashed horizontal arrows** between each matching pair of layers labelled "peer protocol". Down the left side, show the data unit growing:

Layer 7-5		DATA		← PDU: Data
Layer 4		[TCP hdr] DATA		← PDU: Segment (TCP) / Datagram (UDP)
Layer 3		[IP hdr][TCP hdr] DATA		← PDU: Packet
Layer 2		[Frame hdr][IP][TCP] DATA [Trailer]		← PDU: Frame
Layer 1		1010110010110101110101...		← PDU: Bits

That encapsulation strip is often worth 4–5 marks on its own. Draw it every time.

The seven layers in detail

Layer 1 — Physical

Job: transmit **raw bits** over a physical medium. It has no idea what the bits mean.

Aspect	Detail
PDU	Bits
Functions	Bit representation (encoding: NRZ, Manchester); data rate (bit duration); bit synchronisation (sender and receiver clocks); transmission mode (simplex/half/full duplex); physical topology ; line configuration (point-to-point vs multipoint); mechanical and electrical specifications (voltage levels, pin layout, cable and connector type)
Devices	Repeater, hub , cables, connectors, NIC (physical part), modem , transceiver
Standards/ protocols	RS-232, V.35, X.21, RJ-45, DSL, SONET/SDH, IEEE 802.3 physical, Bluetooth PHY

Address	None
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Layer 2 — Data Link

Job: turn the unreliable raw bit pipe into a **reliable link between two directly connected nodes** — hop-to-hop, node-to-node delivery.

Aspect	Detail
PDU	Frame
Functions	Framing (marking where a frame begins and ends); physical (MAC) addressing ; flow control (stop-and-wait, sliding window — preventing a fast sender from swamping a slow receiver); error control — detection via CRC/checksum/parity and correction via retransmission (ARQ) ; access control — deciding who may transmit on a shared medium
Sub-layers	LLC (Logical Link Control, IEEE 802.2) — interfaces to Layer 3, flow/error control. MAC (Media Access Control) — physical addressing and channel access (CSMA/CD, CSMA/CA, token passing)
Devices	Switch , bridge, NIC, wireless access point
Protocols	Ethernet (802.3), Wi-Fi (802.11), PPP, HDLC, SLIP, Frame Relay, ATM, Token Ring (802.5), ARP (sits between 2 and 3)
Address	MAC address (48-bit physical address)

Layer 3 — Network

Job: deliver a packet **end to end, across multiple networks** — source host to destination host, possibly through many intermediate routers. Layer 2 gets you across one hop; Layer 3 gets you across the internetwork.

Aspect	Detail
PDU	Packet (or datagram)
Functions	Logical addressing (IP); routing — choosing the path (routing tables, routing algorithms); forwarding ; fragmentation and reassembly (when the next link's MTU is too small); congestion control ; internetworking; TTL/hop-count management; Quality of Service
Devices	Router , Layer-3 switch
Protocols	IP (IPv4/IPv6), ICMP (error reporting — ping , traceroute), IGMP , ARP/RARP , routing protocols RIP , OSPF , EIGRP , BGP , IPSec
Address	IP address (logical)

Layer 4 — Transport

Job: deliver data reliably (or not) **process to process**. Layer 3 gets a packet to the right *host*; Layer 4 gets it to the right *program on that host*, using **port numbers**. This is the boundary layer — **the first truly end-to-end layer**, and the lowest layer that intermediate routers do not look at.

Aspect	Detail
PDU	Segment (TCP) / Datagram (UDP)

Functions	Service-point (port) addressing ; segmentation and reassembly (with sequence numbers); connection control (connection-oriented vs connectionless); end-to-end flow control (sliding window); end-to-end error control (checksum, acknowledgement, retransmission); multiplexing/demultiplexing several conversations onto one host
Devices	Gateway, some firewalls, load balancers
Protocols	TCP, UDP , SCTP, DCCP
Address	Port number (16-bit)

Layer 5 — Session

Job: establish, manage and terminate the **dialogue** between two applications.

Aspect	Detail
PDU	Data
Functions	Session establishment, maintenance and termination ; dialogue control — deciding whose turn it is (half-duplex or full-duplex conversation); synchronisation — inserting checkpoints into a long transfer so that after a crash you resume from the last checkpoint rather than restarting (the classic example: a 2000-page file transfer with a checkpoint every 100 pages); token management
Protocols	NetBIOS, RPC, PPTP, SQL sessions, NFS, SAP, session part of SIP

Layer 6 — Presentation

Job: deal with the **syntax and semantics** of the data — make sure what the sender means is what the receiver understands. Sometimes called the **translation layer**.

Aspect	Detail
PDU	Data
Functions	Translation between different data representations (ASCII ↔ EBCDIC, big-endian ↔ little-endian, into a common transfer syntax); encryption and decryption for confidentiality; compression and decompression to reduce the bits transmitted
Protocols/formats	SSL/TLS (commonly placed here), MIME, JPEG, GIF, MPEG, ASCII, EBCDIC, XDR, ASN.1

Encryption and compression are the Presentation layer's job — the single most examined fact about Layer 6.

Layer 7 — Application

Job: provide the **interface through which user applications access network services**.

Note carefully: **the layer is not the application itself**. Chrome is not Layer 7; the **HTTP protocol** Chrome speaks is.

Aspect	Detail
PDU	Data / message
Functions	Network virtual terminal (Telnet); file transfer, access and management (FTP); mail services (SMTP); directory services (DNS, LDAP); web browsing (HTTP); network management (SNMP)
Protocols	HTTP/HTTPS, FTP, TFTP, SMTP, POP3, IMAP, DNS, DHCP, SNMP, Telnet, SSH, NFS, LDAP

The consolidated OSI table — reproduce this from memory

#	Layer	PDU	Address	Key functions	Devices	Protocols
7	Application	Data	—	User interface to network services	Gateway, proxy	HTTP, FTP, SMTP, DNS, DHCP, Telnet, SNMP, SSH
6	Presentation	Data	—	Translation, encryption, compression	Gateway	SSL/TLS, MIME, JPEG, MPEG, ASCII
5	Session	Data	—	Session setup/manage/teardown, dialogue control, synchronisation (checkpoints)	Gateway	NetBIOS, RPC, PPTP, SQL
4	Transport	Segment/Datagram	Port	End-to-end delivery, segmentation, flow & error control, connection control	Firewall, gateway	TCP, UDP, SCTP
3	Network	Packet	IP address	Logical addressing, routing, fragmentation, congestion control	Router	IP, ICMP, IGMP, ARP, RIP, OSPF, BGP
2	Data Link	Frame	MAC address	Framing, physical addressing, error detection (CRC), flow control, media access	Switch, bridge, NIC	Ethernet, PPP, HDLC, 802.11, Frame Relay
1	Physical	Bits	—	Bit transmission, encoding, data rate, synchronisation, topology, media specs	Hub, repeater, cable, modem	RS-232, V.35, DSL, SONET, 802.3 PHY

Layer grouping: Layers 1–3 (sometimes 1–4) are the **network support / media layers** — they move data. Layers 5–7 are the **user support / host layers** — they make it meaningful. Layer 4 is the **bridge** between the two.

Merits and demerits of OSI

Advantages	Disadvantages
Standardised — vendor-independent interoperability	Never fully implemented — TCP/IP won in practice
Modular — a change in one layer doesn't force changes elsewhere	Layers 5 and 6 are thin/underused , layers 2 and 3 overloaded
Simplifies teaching, design and troubleshooting (isolate faults to a layer)	The model was defined before the protocols , so it fits nothing exactly
Supports both connection-oriented and connectionless service at the network layer	Complex ; overhead of headers at every layer
Clear division of labour; encourages competition at each layer	Some functions (addressing, flow control, error control) are duplicated across layers

Likely exam questions

- [5] What is the OSI model? Name the seven layers in order.

- **[5]** Explain the functions of the Data Link layer.
- **[5]** Explain the functions of the Transport layer. Name its PDU and its protocols.
- **[5]** What is encapsulation? Explain with the PDU at each layer.
- **[5]** At which layer does encryption take place? Justify.
- **[10]** Explain the OSI reference model with a neat diagram, describing the function of each of the seven layers.
- **[15]** Describe the OSI reference model in full detail. For each layer, state its function, its PDU, the addresses it uses, the devices that operate there and its representative protocols. Explain encapsulation with a diagram, and discuss the merits and demerits of the model.

MCQ traps

Trap	Correct answer
"How many layers in OSI? Who developed it?"	7, ISO, in 1984.
"PDU at the Network layer?"	Packet. Transport = segment , Data Link = frame , Physical = bits .
"Which layer does routing?"	Network (3).
"Which layer does error <i>detection</i> using CRC?"	Data Link (2). End-to-end error control → Transport (4).
"Encryption/compression layer?"	Presentation (6).
"Dialogue control and checkpoints?"	Session (5).
"Which is the first end-to-end layer?"	Transport (4).
"Which layer does flow control?"	Both 2 (hop) and 4 (end-to-end) — read the question carefully.
"Which layer has no header?"	Physical (it transmits bits).
"Fragmentation is done at?"	Network layer , by the router.
"Is the web browser layer 7?"	The protocol (HTTP) is; the application program sits <i>above</i> the model.

8. The TCP/IP model

Concept

TCP/IP came first (ARPANET, 1970s) and OSI came later as a tidy abstraction. That is why TCP/IP is untidy but real: it has **four layers** (some texts say five, splitting Network Access into Physical and Data Link), and its layers were **derived from working protocols** rather than designed in advance.

TCP/IP layer	Maps to OSI	Function	Protocols
Application	7 + 6 + 5	All host-level functions: user services, translation, session	HTTP, FTP, SMTP, POP3, IMAP, DNS, DHCP, Telnet, SSH, SNMP, TFTP
Transport (Host-to-Host)	4	End-to-end delivery, ports, reliability	TCP, UDP
Internet	3	Logical addressing and routing	IP , ICMP, IGMP, ARP, RARP

Network Access (Link / Network Interface)	2 + 1	Framing, MAC addressing, physical transmission	Ethernet, Wi-Fi, PPP, Frame Relay, ATM, token ring
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OSI vs TCP/IP — the comparison table

Criterion	OSI	TCP/IP
Number of layers	7	4 (or 5)
Developed by	ISO	DARPA / DoD (US Department of Defense)
Order of creation	Model first, protocols after	Protocols first, model after
Nature	A reference/theoretical model	An implemented, practical protocol suite
Layer independence	Strictly independent; well-defined interfaces	Layers less strictly separated
Network layer service	Both connection-oriented and connectionless	Connectionless only (IP)
Transport layer service	Connection-oriented only	Both — TCP (connection-oriented) and UDP (connectionless)
Session and Presentation	Separate layers (5, 6)	Absent — merged into Application
Protocol dependence	Protocol-independent — a generic standard	Protocol-dependent — built around TCP and IP
Usage today	Teaching, reference, troubleshooting vocabulary	The actual Internet
Reliability of the base	Guaranteed at multiple layers	Reliability is the Transport layer's job; IP is best-effort

Common port numbers — pure MCQ material

Port	Protocol	Transport
20 / 21	FTP (20 data, 21 control)	TCP
22	SSH / SFTP / SCP	TCP
23	Telnet	TCP
25	SMTP	TCP
53	DNS	UDP (queries) and TCP (zone transfers, large responses)
67 / 68	DHCP (67 server, 68 client)	UDP
69	TFTP	UDP
80	HTTP	TCP
110	POP3	TCP
119	NNTP	TCP

143	IMAP	TCP
161 / 162	SNMP / SNMP trap	UDP
443	HTTPS	TCP
445	SMB	TCP
465 / 587	SMTPS / SMTP submission	TCP
993 / 995	IMAPS / POP3S	TCP
3306 / 1521 / 5432	MySQL / Oracle / PostgreSQL	TCP
3389	RDP	TCP

Port ranges: **0–1023 well-known**, **1024–49151 registered**, **49152–65535 dynamic/private (ephemeral)**.

MCQ traps

Trap	Correct answer
"How many layers in TCP/IP?"	4 (Application, Transport, Internet, Network Access).
"Which OSI layers are missing from TCP/IP?"	Session and Presentation (absorbed into Application).
"Which model came first?"	TCP/IP protocols ; OSI model was standardised in 1984.
"DNS uses which transport?"	UDP port 53 normally, TCP for zone transfers.
"Port 25 / 110 / 143?"	SMTP / POP3 / IMAP.
"TCP/IP internet layer is connectionless?"	Yes — IP is best-effort.

11. The Transport layer — TCP and UDP

Concept

IP is **best-effort**: it will try to deliver your packet, but it makes no promise about arrival, order, duplication or corruption. Somebody has to decide whether that is acceptable. That is the Transport layer's choice, and it offers exactly two answers:

- **TCP** — "I will make it look like a reliable, ordered byte stream, whatever it costs."
- **UDP** — "I will add ports and a checksum and get out of the way."

Neither is better; they are different trade-offs of **reliability against latency and overhead**.

TCP vs UDP — the table

Criterion	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented — handshake before data	Connectionless — just send
Reliability	Reliable — acknowledgements + retransmission	Unreliable — fire and forget

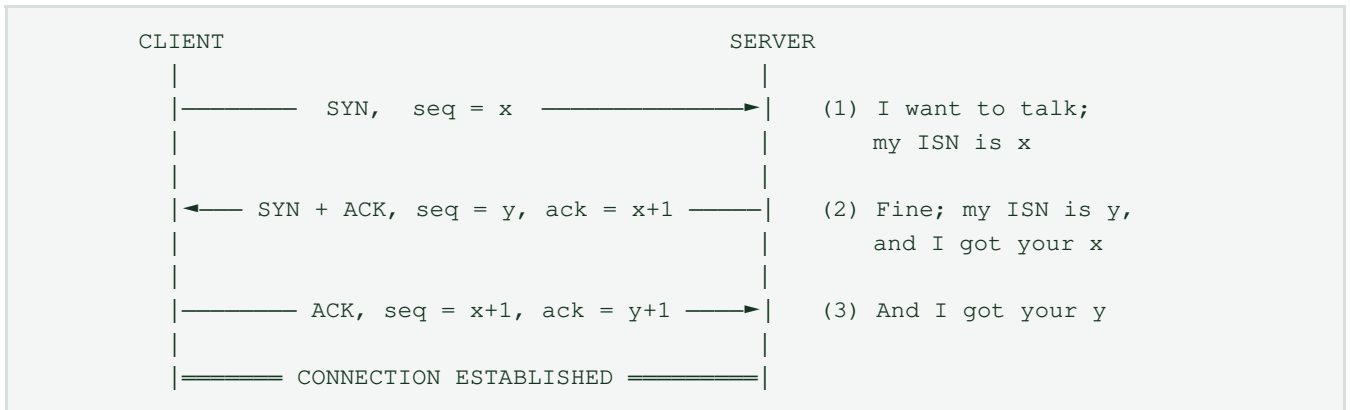
Ordering	Guaranteed in-order delivery (sequence numbers)	No ordering guarantee
Duplicate detection	Yes	No
Error checking	Checksum + recovery	Checksum only, discards bad datagrams
Flow control	Yes — sliding window, receiver-advertised	No
Congestion control	Yes — slow start, congestion avoidance, fast retransmit/recovery	No
Header size	20 bytes minimum (up to 60 with options)	8 bytes fixed
Speed / overhead	Slower, heavy	Fast, lightweight
Data unit (PDU)	Segment	Datagram
Transmission style	Byte stream (no message boundaries preserved)	Message/datagram oriented (boundaries preserved)
Broadcast/multicast	Not supported (point-to-point only)	Supported
Applications	HTTP/HTTPS, FTP, SMTP, POP3, IMAP, SSH, Telnet — anything where correctness matters	DNS, DHCP, TFTP, SNMP, VoIP, video streaming, online gaming, RIP — anything where speed beats perfection

UDP's 8-byte header has exactly four 2-byte fields: **source port, destination port, length, checksum**. Being able to name all four is an easy mark.

TCP header key fields (20 bytes): source port, destination port, **sequence number (32 bits), acknowledgement number (32 bits)**, header length/offset, **flags (URG, ACK, PSH, RST, SYN, FIN)**, **window size**, checksum, urgent pointer, options.

The three-way handshake — draw this

TCP must synchronise sequence numbers in **both** directions before data flows. Two messages would establish it one way only; three is the minimum for both.



Step	Segment	Flags	Meaning
1	Client → Server	SYN , seq = x	Client asks to open, sends its Initial Sequence Number x

2	Server → Client	SYN + ACK , seq = y, ack = x+1	Server agrees, sends its own ISN y, and acknowledges x
3	Client → Server	ACK , ack = y+1	Client acknowledges y. Connection is now ESTABLISHED

Note the acknowledgement number is always **the next byte expected**, hence x+1 not x. The SYN flag itself consumes one sequence number.

Connection termination is a FOUR-way handshake (FIN → ACK → FIN → ACK), because TCP connections are **full-duplex** and each direction must be closed independently — one side can finish sending while still receiving ("half-close"). Confusing three-way (setup) with four-way (teardown) is a very common MCQ trap.

SYN flood attack (relevant to your Forensic paper): the attacker sends thousands of SYNs with spoofed source addresses and never sends the third ACK. Each half-open connection consumes a slot in the server's backlog queue until legitimate clients cannot connect — a **denial of service**. Mitigation: **SYN cookies**, reduced timeouts, rate limiting, firewalls.

Flow control and congestion control — distinguish them

	Flow control	Congestion control
Protects	The receiver from being overwhelmed	The network from being overwhelmed
Mechanism	Sliding window , receiver advertises rwnd	Sender maintains cwnd : slow start (exponential growth), congestion avoidance (linear, after the threshold), fast retransmit on 3 duplicate ACKs, fast recovery
Scope	End to end, two parties	The whole path

Sender's usable window = **min(rwnd, cwnd)**.

Error control mechanisms to name: **checksum** for detection; **acknowledgement** (positive, cumulative); **retransmission on timeout** (RTO, computed from a smoothed round-trip-time estimate); ARQ schemes — **stop-and-wait**, **go-back-N**, **selective repeat**.

Likely exam questions

- **[5]** Differentiate between TCP and UDP.
- **[5]** Explain the TCP three-way handshake with a diagram.
- **[5]** Why is TCP connection termination a four-way handshake?
- **[5]** Distinguish between flow control and congestion control.
- **[5]** Name the fields of the UDP header. Why is UDP used for DNS and VoIP?
- **[10]** Explain the functions of the Transport layer. Compare TCP and UDP in detail and explain connection establishment and termination with diagrams.
- **[15]** Explain TCP in detail: header format, three-way handshake, sliding-window flow control, congestion control (slow start and congestion avoidance) and error recovery. Contrast with UDP and state where each is appropriate.

MCQ traps

Trap	Correct answer
"TCP header minimum size?"	20 bytes . UDP = 8 bytes .

"Handshake steps for setup / teardown?"	3 for setup, 4 for teardown.
"Which flags in step 2?"	SYN + ACK .
"Ack number in step 2?"	x + 1 , not x.
"Which supports multicast?"	UDP .
"DNS uses?"	UDP (53) normally.
"Does UDP have a checksum?"	Yes — but it discards bad datagrams; no recovery.
"TCP preserves message boundaries?"	No — it is a byte stream . UDP does.